Atividade: Atribuir variáveis em Python

Introdução

As variáveis ajudam os analistas de segurança a monitorar uma variedade de informações relacionadas à segurança. Por exemplo, os analistas podem precisar criar variáveis em Python para os usuários que têm permissão para fazer login, o número de tentativas de login permitidas para cada usuário e o número atual de tentativas realizadas por um usuário.

Neste laboratório, você praticará a atribuição de valores a variáveis e a determinação de seus tipos de dados.

Dicas para concluir este laboratório

Cenário

Você é um analista de segurança responsável por escrever um código que automatizará a análise de tentativas de login em um dispositivo específico. Como primeiro passo, você precisará criar variáveis para armazenar informações relevantes ao processo de login. Essas informações incluem o ID do dispositivo, a lista de nomes de usuário aprovados, o número máximo de tentativas de login permitidas por usuário, as tentativas de login realizadas pelo usuário e o status do login.

Ao longo deste laboratório, você atribuirá essas variáveis e verificará os tipos de dados das variáveis.

Tarefa 1

Em seu trabalho como analista, imagine que existe um dispositivo ao qual apenas os usuários especificados em uma lista de permissões podem acessar, e cujo ID do dispositivo é `"72e08x0"` .

Na célula de código a seguir, atribua esse valor a uma variável chamada `device_id` . Em seguida, exiba o conteúdo da variável e observe a saída.

Certifique-se de substituir cada um `### YOUR CODE HERE ###` pelo seu próprio código antes de executar a próxima célula.

Em [1]:
```
# Atribua à variável `device_id` o ID do dispositivo que somente usuários e específicos podem acessar.

device_id  =  "72e08x0"

# Exibir `device_id`

imprimir ( id_do_dispositivo )
```

```
72e08x0
```

Dica 1

Dica 2

Dica 3

Tarefa 2

Agora que a variável `device_id` está definida, você pode retornar seu tipo de dados.

In this task, use a Python function to find the data type of the variable `device_id` . Store the data type in another variable called `device_id_type` . Then, display `device_id_type` to examine the output.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [4]:
```
# Assign the `device_id` variable to the device ID that only specified users can access

device_id = "72e08x0"

# Assign `device_id_type` to the data type of `device_id`

device_id_type = type(device_id)

# Display `device_id_type`

print(device_id_type)
```

```
<class 'str'>
```

Hint 1

Hint 2

Hint 3

Question 1

Based on the output above, what do you observe about the data type of `device_id` ?

Eu observo que o tipo de dados da variável device_id é uma string (representada por str).

Task 3

As you continue your work, you're provided a list of usernames of users who are allowed to access the device. The usernames with this access are `"madebowa"` , `"jnguyen"` , `"tbecker"` , `"nhersh"` , and `"redwards"` .

In this task, create a variable called `username_list` . Assign a list with the approved usernames to this variable. Then, display the value of the `username_list` variable.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [5]:
```
# Assign `username_list` to the list of usernames who are allowed to access
the device

username_list = ["madebowa", "jnguyen", "tbecker", "nhersh", "redwards"]

# Display `username_list`

print(username_list)
```

```
['madebowa', 'jnguyen', 'tbecker', 'nhersh', 'redwards']
```

Hint 1

Hint 2

Hint 3

Task 4

In this task, find the data type of the `username_list` . Store the type in a variable called `username_list_type` . Then, display `username_list_type` to examine the output.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [7]:
```
# Assign `username_list` to the list of usernames who are allowed to access the device

username_list = ["madebowa", "jnguyen", "tbecker", "nhersh", "redwards"]

# Assign `username_list_type` to the data type of `username_list`

username_list_type = type(username_list)

# Display `username_list_type`

print(username_list_type)
```

```
<class 'list'>
```

Hint 1

Hint 2

Hint 3

Question 2

Based on the output above, what do you observe about the data type of `username_list` ?

tipo dados de lista

Task 5

Now, imagine that you've been informed that the previous list is not up-to-date and that there is another employee that now has access to the device. You're given the updated list of usernames with access, including the new employee, as follows: `"madebowa"` , `"jnguyen"` , `"tbecker"` , `"nhersh"` , `"redwards"` , and `"lpope"` .

In this task, reassign the variable `username_list` to the new list. Run the code to display the list before and after it's been updated to observe the difference.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [8]:
```
# Assign `username_list` to the list of usernames who are allowed to access the device

username_list = ["madebowa", "jnguyen", "tbecker", "nhersh", "redwards"]

# Display `username_list`

print(username_list)

# Assign `username_list` to the updated list of usernames who are allowed to access the device

username_list = ["madebowa", "jnguyen", "tbecker", "nhersh", "redwards", "lpope"]

# Display `username_list`

print(username_list)
```

```
['madebowa', 'jnguyen', 'tbecker', 'nhersh', 'redwards']
['madebowa', 'jnguyen', 'tbecker', 'nhersh', 'redwards', 'lpope']
```

Hint 1

Hint 2

Question 3

Based on the output above, what do you observe about the contents of `username_list` ?

Lista de funcionários atualizou.

Task 6

In this task, define a variable called `max_logins` that represents the maximum number of login attempts allowed per user. Store the value `3` in this variable. Then, store its data type in another variable called `max_logins_type` . Display `max_logins_type` to examine the output.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [9]:
```
# Assign `max_logins` to the value 3

max_logins = 3

# Assign `max_logins_type` to the data type of `max_logins`

max_logins_type = type(max_logins)

# Display `max_logins_type`

print(max_logins_type)
```

```
<class 'int'>
```

Hint 1

Hint 2

Hint 3

Question 4

Based on the output above, what do you observe about the data type of `max_logins` ?

O output <class 'int'> confirma que o Python identificou o valor 3 como um número inteiro.

Task 7

In this task, define a variable called `login_attempts` that represents the current number of login attempts made by a user. Store the value `2` in this variable. Then, store its data type in a variable called `login_attempts_type` . Display `login_attempts_type` to observe the output.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [10]:
```
# Assign `login_attempts` to the value 2

login_attempts = 2

# Assign `login_attempts_type` to the data type of `login_attempts`

login_attempts_type = type(login_attempts)

# Display `login_attempts_type`

print(login_attempts_type)
```

```
<class 'int'>
```

Hint 1

Hint 2

Hint 3

Question 5

Based on the output above, what do you observe about the data type of `login_attempts` ?

O output <class 'int'> indica que a variável login_attempts é um número inteiro.

Task 8

In this task, you'll determine the Boolean value that represents whether the current number of login attempts a user has made is less than or equal to the maximum number of login attempts allowed.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [11]:
```
# Assign `max_logins` to the value 3

max_logins = 3

# Assign `login_attempts` to the value 2

login_attempts = 2

# Determine whether the current number of login attempts a user has made is
less than or equal to the maximum number of login attempts allowed,
# and display the resulting Boolean value

print(login_attempts <= max_logins)
```

```
True
```

Hint 1

Hint 2

Question 6

What is the output? What does this mean?

True, que as lentativas de login do login_attempts são menores que do max_logins

Task 9

This code continues to check for the Boolean value of whether `max_logins` is less than or equal to `login_attempts`. In this task, reassign other values to `login_attempts`. For example, you might choose a value that is higher than the maximum number of attempts allowed. Observe how the output changes.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [12]:
```
# Assign `max_logins` to the value 3

max_logins = 3

# Assign `login_attempts` to a specific value

login_attempts = 5

# Determine whether the current number of login attempts a user has made is
less than or equal to the maximum number of login attempts allowed,
# and display the resulting Boolean value

print(login_attempts <= max_logins)
```

```
False
```

Hint 1

Question 7

Based on the different values you assigned to `login_attempts`, what did you observe about the output?

Com o aumento da tentivas no login_attempts o resultado mundo para "False". afirmando que max_logins agora tem menos tentaivas que login_attempts.

Task 10

Finally, you can also assign a Boolean value of `True` or `False` to a variable.

In this task, you'll create a variable called `login_status`, which is a Boolean that represents whether a user is logged in. Assign `False` to this variable and store its data type in a variable called `login_status_type` and display it.

Be sure to replace each `### YOUR CODE HERE ###` with your own code before you run the following cell.

In [13]:
```
# Assign `login_status` to the Boolean value False

login_status = False

# Assign `login_status_type` to the data type of `login_status`

login_status_type = type(login_status)

# Display `login_status_type`

print(login_status_type)
```

```
<class 'bool'>
```

Hint 1

Hint 2

Question 8

Based on the output above, what do you observe about the data type of `login_status` ?

O output <class 'bool'> indica que a variável login_status é do tipo booleano. Isso significa que ela armazena um valor lógico, que no Python só pode ser True (Verdadeiro) ou False (Falso).

Conclusion

What are your key takeaways from this lab?

Aprendi a atribuir valores a variáveis, identificar tipos de dados (str, int, list, bool) com a função type() e usar operadores lógicos para comparar informações.